
SKILLS

- **AI/LLM Systems:** LLMs, RAG, LangChain, LangGraph, Agentic AI, tool calling, embeddings, OpenAI API
- **Vector/Retrieval:** pgvector, Pinecone, HuggingFace, context assembly, grounding controls, citation validation
- **LLM Evaluation:** offline eval sets, regression metrics, prompt comparison, answer quality, guardrails, tracing
- **Full Stack:** Python, TypeScript, JavaScript, Go, SQL, React, Node.js, Express, REST APIs, GraphQL
- **Distributed Systems:** microservices, API design, PostgreSQL, Redis, Kafka, RabbitMQ, event streams
- **Data Platforms:** Snowflake, Airflow, Elasticsearch, schema design, caching, analytics pipelines, vector search
- **Cloud/Infra:** AWS Bedrock, SageMaker, Lambda, EKS, Kinesis, S3, Docker, Kubernetes, Terraform, CI/CD
- **Reliability/Ops:** OpenTelemetry, ELK, CloudWatch, SLOs, alerting, production debugging, incident response

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect – Applied AI / Full Stack Nov 2022 – Present
 - **ASB Assist – Agentic GenAI Platform:** Building conversational AI on RBC's enterprise blueprint platform: natural-language requirements in, compliant architecture diagrams and service recommendations out; LLM + RAG pipeline with LangChain orchestration, tool integration, prompt/context engineering, audit trails, and iterative offline evaluation of answer quality and safety.
 - **RAG & Evaluation Infrastructure:** Designed retrieval patterns, grounding controls, context assembly, and model-evaluation workflows for regulated architecture use cases; compare prompt versions, detect regressions, validate citations, and track quality before production rollout.
 - **0-to-1 Full-Stack Platform Delivery:** Led production delivery of patent-pending blueprint platform (Node.js/TypeScript, React) with rules-based recommendation engine, real-time data rendering, auto-certification, and secure service matching; wrote 99.6% of production code and 100% of core logic.
 - **High-Scale Observability Platform:** Architected telemetry platform using OpenTelemetry, Vector, Fluent Bit, Kafka, and Logstash ingesting 150 TB/day across 10,000+ systems; standardised schemas enabling AIOps, distributed tracing, latency SLOs, alerting, and production debugging.
 - **Technical Leadership & Production Ownership:** Led 5 engineers through technical planning, execution, reviews, ADRs, runbooks, operational handoff, and stakeholder alignment with product, risk, security, infrastructure, and architecture teams.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **AI-Enabled Healthcare/Insurance Platform:** Joined as first engineer and built React/TypeScript frontend, Node.js/Express APIs, Hyperledger Fabric verification, and AWS EKS deployment with HIPAA audit logging; team scaled to 20+ engineers before acquisition.
 - **Production ML & Distributed Services:** Integrated transformer-based document classification and trust-score workflows for claims/credential verification; deployed inference APIs, monitoring, and audit trails, reducing manual review by 60%.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
Software Developer Co-op Apr 2022 – Nov 2022
 - **Modern Web Platform:** Migrated federal tax platform UI from Apache Struts to React/TypeScript; contributed to backend API layers serving millions of Canadians with WCAG accessibility, automated quality gates, and maintainable code standards.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Systems Research & Cloud Teaching:** Researched call-stack decision algorithms and Python tracing tools for compiler/runtime analysis; TA for Algorithms, Software Development, and graduate Advanced Cloud Computing.

PROJECTS

- **Enterprise Document Q&A with LLM Eval Harness:** LangChain + pgvector RAG over messy document corpora with offline eval sets, regression metrics, prompt/version comparison, citation validation, and tracing for production-style answer-quality iteration.
- **Automatic Trading Algorithm:** Sub-second latency personal project integrating live brokerage APIs and real-time market streams; momentum/mean-reversion signals with automated risk management, position sizing, and performance monitoring.
- **Flight Price Optimization Agent:** DQN/DDQN reinforcement-learning agent in Python for dynamic pricing; modelled demand curves on historical flight data and improved simulated outcome by 18% over baseline.

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020